

We can then assign a label from the infinite Ulam-Harris tree $\mathcal{I} := \bigcup_{n \in \mathbb{N}_0} \mathbb{N}^n$ to each individual $u = (u_1, \dots, u_n) = u_1 \dots u_n \in \mathcal{I}$ according to its ancestral line, namely,

$$\emptyset \rightarrow u_1 \rightarrow u_1 u_2 \rightarrow \dots \rightarrow u_1 \dots u_n = u,$$

where u_1 is the u_1 -th child of the ancestor, $u_1 u_2$ is the u_2 -th child of u_1 , and so on. We write $|u| = n$ if $u \in \mathbb{N}^n$.

Now let $\xi_u = (\xi_u^{i,j})_{i,j=1,\dots,p}$, $u \in \mathcal{I} \setminus \{\emptyset\}$ be a family of i.i.d. copies of $\xi =: \xi_\emptyset$. We suppose that $(\Omega, \mathcal{A}, \mathbf{P})$ is a probability space on which all ξ_u , $u \in \mathcal{I}$ are defined and i.i.d. and independent of $\tau(\emptyset)$, the random variable that gives the ancestor's type. We write $\mathbf{P}^i$ if the ancestor's type is $i \in [p]$. We denote the associated expected value operators by $\mathbf{E}$ and $\mathbf{E}^i$, respectively.

Each individual $u \in \mathcal{T}$ that is ever born has a clearly defined type $\tau(u) \in [p]$ and a time of birth $S(u)$, namely, $S(\emptyset) = 0$ and, recursively,

$$S(uk) = S(u) + X_{u,k}, \quad u \in \mathcal{I}, k \in \mathbb{N}.$$

For completeness, we write $S(u) = \infty$ if individual u is never born.

Now suppose that for each $u \in \mathcal{I}$ there exist a random vector $\zeta_u = (\zeta_u^1, \dots, \zeta_u^p)^\top$ taking values in $[0, \infty]^p$ and a product-measurable, separable random characteristic $\varphi_u = (\varphi_u^1, \dots, \varphi_u^p)^\top : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^p$ (taking values in the space of p -dimensional real column vectors) such that the (ξ_u, ζ_u) , $u \in \mathcal{I}$ are i.i.d. and the φ_u , $u \in \mathcal{I}$ are identically distributed. Further, we assume throughout that for every $n \in \mathbb{N}$, the φ_u , $u \in \mathbb{N}^n$ are independent and independent of the $(\xi_u, \zeta_u, \varphi_u)$, $|u| < n$. These assumptions are satisfied, for instance, when the $(\xi_u, \zeta_u, \varphi_u)$, $u \in \mathcal{I}$ are i.i.d. However, our assumption is weaker and allows φ_u to be a function of the (ξ_{uv}, ζ_{uv}) , $v \in \mathcal{I}$. The variable ζ_u^i is viewed as the lifetime of the potential individual u given that its type is i . The function $\varphi_u^i(t)$ is some kind of score assigned to the individual u at the age t given that its type is i . Hence, for $u \in \mathcal{T}$, we define $\zeta_u := \zeta_u^{\tau(u)}$ as the lifetime of individual u and $\varphi_u(t) := \varphi_u^{\tau(u)}(t) = \mathbf{e}_{\tau(u)}^\top \varphi_u(t)$ as the score assigned to individual u at the age t . For $t \geq 0$ and $j \in [p]$, define

$$Z_t^j := \sum_{u \in \mathcal{T}} \mathbb{1}_{\{j\}}(\tau(u)) \mathbb{1}_{[0, \zeta_u)}(t - S(u)),$$

i.e., Z_t^j is the number of individuals of type j alive at time t . Now write

$$Z_t := (Z_t^1, \dots, Z_t^p)$$

for the associated row vector. Then $(Z_t)_{t \geq 0}$ is the multi-type general branching process. We are interested in the asymptotic behavior of $\mathbf{E}^i[Z_t]$ as $t \rightarrow \infty$ for $i = 1, \dots, p$. More generally, we are interested in the asymptotic behavior of $\mathbf{E}^i[Z_t^\varphi]$ as $t \rightarrow \infty$ for the multi-type general branching process counted with characteristic φ defined via

$$Z_t^\varphi := \sum_{u \in \mathcal{T}} \varphi_u(t - S(u)), \quad t \in \mathbb{R}.$$

Notice that introducing a general score φ indeed generalizes the model since we may write $Z_t^j = Z_t^\varphi$ for φ_u with

$$\varphi_u^i(t) = \begin{cases} 0 & \text{for } i \neq j, \\ \mathbb{1}_{[0, \zeta_u^j)} & \text{for } i = j. \end{cases}$$